

Chemistry

Unit 1

Practice Exam 1

This sample exam paper has been prepared as part of the Pearson suite of resources for the Year 11, Unit 1, ATAR Chemistry Course prescribed by the Western Australian School Curriculum and Standards Authority.

Time allowed

Reading time: 10 minutes Working time: 90 minutes

Materials required

An approved non-programmable calculator

Chemistry Data Sheet. This may be downloaded from the WACE website.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section one: Multiple-choice	13	13	25	13	25
Section two: Short answer	7	7	35	38	40
Section three: Extended answer	3	3	30	37	35
				Total	100

Section one: Multiple-choice

25% (13 marks)

This section has 13 questions. Answer **all** questions by circling the correct option.

Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers.

No marks will be given if more than one answer is completed for any question.

Suggested working time: 25 minutes

1 Consider the following electrostatic forces.

- I Attraction between oppositely charged particles
- II Repulsion between positively charged particles
- III Repulsion between negatively charged particles

The electrostatic forces operating in an atom are

- a I only.
- b I and II only.
- c I and III only.
- d I, II and III.

2 Element X is in group 4 of the periodic table. Element Y is in group 16. Identify the most likely formula for a compound between X and Y.

- a X_4Y_{16}
- b $X_{16}Y_4$
- c XY_4
- d XY_2

3 Zinc is an element. Therefore,

- a zinc has no isotopes.
- b all zinc atoms are identical.
- c all zinc atoms contain the same number of protons.
- d zinc atoms contain equal numbers of protons and neutrons.

4 Which pair of elements are chemically most similar to each other?

- a Li and Be
- b Cl and Ar
- c N and P
- d Ne and Na

5 Copper has two isotopes, ^{63}Cu and ^{65}Cu . It has a relative atomic mass of 63.55. So, any one particular copper atom would

- a contain either 63 or 65 protons.
- b contain either 34 or 36 neutrons.
- c have a mass number of 63.55.
- d have an atomic number of 29 or 31.

- 6 The empirical formula of a molecular compound is best defined as
- the actual number and type of each element present.
 - the simplest whole number mole ratio of each ion present.
 - the simplest whole number mole ratio of each element present.
 - the simplest whole number mass ratio of each element present.
- 7 The electron configuration of four different elements is given below. Which one is most likely to form a stable ion with a valency of -3 ?
- 2,5
 - 2,8,2
 - 2,8,3
 - 2,8,8
- 8 In the molten state, an ionic compound conducts electricity because
- ions are free to move.
 - atoms are free to move.
 - electrons are free to move.
 - molecules are free to move.
- 9 Predict which one of the following ionic compounds will have the highest melting point.
- MgCl_2
 - MgO
 - Na_2O
 - NaCl
- 10 Identify the correct formula for sodium sulfate.
- NaS
 - Na_2S
 - NaSO_4
 - Na_2SO_4
- 11 Butane (C_4H_{10}) reacts with oxygen (O_2) according to the following equation:
- $$\text{C}_4\text{H}_{10}(\text{g}) + \frac{13}{2}\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 5\text{H}_2\text{O}(\text{g}) \quad \Delta H = -2877 \text{ kJ mol}^{-1}$$
- Which one of the following statements about the information above is correct?
- The energy of the bonds broken in the reactants is **greater** than the energy of the bonds formed in the product and so the reaction is exothermic.
 - The energy of the bonds broken in the reactants is **less** than the energy of the bonds formed in the product and so the reaction is exothermic.
 - The energy of the bonds broken in the reactants is **greater** than the energy of the bonds formed in the product and so the reaction is endothermic.
 - The energy of the bonds broken in the reactants is **less** than the energy of the bonds formed in the product and so the reaction is endothermic.

- 12** 0.25 mol of one of the hydrocarbons listed below has a mass of 19.5 g. Select the correct hydrocarbon.
- a** CH_4
 - b** C_3H_6
 - c** C_6H_6
 - d** C_6H_{14}
- 13** A mass spectrometer can be used to identify the presence of isotopes in a compound. Isotopes are best defined as atoms
- a** that have a higher number of protons than electrons.
 - b** that have a higher number of neutrons than protons.
 - c** with the same number of neutrons and a different number of protons.
 - d** with the same number of protons and a different number of neutrons.

End of section one

Section two: Short answer

40% (38 marks)

This section has 7 questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 35 minutes

Question 14

(4 marks)

- a** Complete the table below by writing the formula of each of the compounds listed. (2 marks)

Name of compound	Formula of compound
Potassium phosphate	
Aluminium oxide	

- b** Complete the table below by writing the name of each of the compounds listed. (2 marks)

Formula of compound	Name of compound
$\text{Ca}(\text{NO}_3)_2$	
CuCO_3	

Question 15

(7 marks)

- a** Magnesium is a commonly used structural metal.

- i** Describe the structure of solid magnesium and the nature of the bonding between its particles. You may include a labelled diagram in your answer. (3 marks)

- ii** Explain why magnesium is a good conductor of electricity. (1 mark)

- b** Use the ionic bonding model to explain the following properties of ionic compounds. (3 marks)

Solid ionic compounds cannot conduct electricity.	
Ionic compounds are generally brittle.	
Ionic compounds have high melting temperatures.	

Question 16

(5 marks)

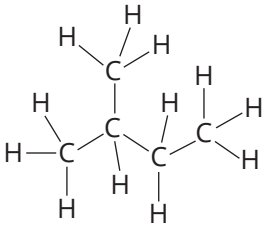
Provide concise explanations for each of the following observations.

- a** When a solution of sodium hydroxide is mixed with a solution of hydrochloric acid, the temperature of the resultant solution increases. (2 marks)

- b** When a sample of a sodium compound is placed in a flame, a yellow colour is observed but when a sample of a potassium compound is placed in a flame, a lilac colour is observed. Your answer should include a reference to the structure of each of these atoms. (3 marks)

Question 17**(5 marks)**

- a** Complete the table below by writing the name of each of the compounds listed. (3 marks)

Formula of compound	Name of compound
$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$	
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	
	

- b** Complete the table below by drawing a structural formula for each of the compounds listed. (2 marks)

Name of compound	Structural formula of compound
hex-1-ene	
2,3-dimethylpentane	

Question 18**(6 marks)**

Write chemical equations for the following reactions.

- a** The addition of bromine to propene. Write structural or semi-structural formulas for all carbon compounds. (2 marks)
- b** The combustion of ethane in excess oxygen. (2 marks)
-
- c** A substitution reaction between benzene and chlorine gas. (2 marks)
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Question 19**(6 marks)**

A sample of aluminium nitrate ($\text{Al}(\text{NO}_3)_3$) has a mass of 30.5 g.

- a** Calculate the amount, in mol, of aluminium nitrate in the sample. (2 marks)

- b** Calculate the amount, in mol, of nitrogen atoms present. (1 mark)

- c** Calculate the total number of atoms present in the sample. (2 marks)

- d** Determine the percentage, by mass, of oxygen in aluminium nitrate. (1 mark)

Question 20**(5 marks)**

A compound found in milk consists of carbon, hydrogen and oxygen only. It contains 54.54% carbon and 36.36% oxygen.

- a** Determine the empirical formula of this compound. (3 marks)

- b** 0.350 mol of this compound has a mass of 30.8 g. Calculate the relative molecular mass of the compound. (1 mark)

- c** Determine the molecular formula of the compound. (1 mark)

End of section two

Section three: Extended answer

35% (37 marks)

This section has 3 questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 30 minutes

Question 21

(13 marks)

a Consider the ionic compound calcium fluoride.

i Write the electron configuration of a calcium atom in its ground state. (1 mark)

ii Write the electron configuration of a calcium ion in its ground state. (1 mark)

iii Write the electron configuration of a fluoride ion in its ground state. (1 mark)

iv What is the ratio of calcium ions to fluoride ions in the compound? (1 mark)

v With reference to the electron configurations of calcium and fluorine, explain your answer to part iv. (1 mark)

b Predict whether the atomic radius of calcium will be greater than or smaller than that of sodium. Justify your prediction. (2 marks)

c Consider the following electron configuration: 2,8,7,1

Explain whether this electron configuration represents a ground state or excited atom. (1 mark)

d Give a brief definition of each of the following terms:

i mass number (1 mark)

ii atomic number (1 mark)

iii isotopes (1 mark)

iv ion (1 mark)

e Give the symbol for the element that forms a -2 ion and has an electron configuration of 2,8. (1 mark)

Question 22 (13 marks)

Consider the melting points of the following substances:

Substance	Melting point ($^{\circ}\text{C}$)
Silicon dioxide	1700
Carbon dioxide	-57
Iron	1540
Strontium fluoride	1400

a Carbon and silicon are in the same group of the periodic table yet they form oxides with widely different melting points. Provide an explanation for this difference in melting point in terms of the structure and bonding of these oxides. You may use a diagram to illustrate your answer. (4 marks)

b Consider the three compounds silicon dioxide, iron and strontium fluoride. These all have very high melting points.

i What does a high melting point indicate about the bonds between the particles in these three compounds? (1 mark)

- ii Compare the structure and bonding in each of these three compounds. (3 marks)

- iii Explain why only one of these three compounds is a conductor of electricity in the solid state. (2 marks)

- iv In which of these three compounds is the bonding electrostatic in nature? (1 mark)

- c Give the formulas of two other compounds with the same type of bonding as strontium fluoride. (2 marks)

Question 23

(11 marks)

The grid below contains formulas for 12 different substances.

$C_6H_{12}O_6$	H_2O	$NaCl$	CH_2
Si	Cu	$CuSO_4$	C_{60}
C_3H_8	C (graphite)	O_2	C_5H_{10}

- a Identify substances from this grid that meet the following descriptions. Note that:
- the same substance may be used more than once
 - a complete answer may require the inclusion of more than one substance. (8 marks)

Select a substance or substances

- i represented by a formula which is both a molecular and an empirical formula.

- ii represented by a formula which is molecular but not empirical.

- iii with a double bond.

- iv with different allotropes.

- v with structural isomers.

- vi that can conduct electricity in the molten form but not as a solid.

vii that contain(s) 14.3% hydrogen.

viii held together by both ionic and covalent bonds.

- b** You are given a solid mixture known to contain only NaCl and C(graphite). Describe how you could obtain a pure sample of NaCl and C(graphite) from this mixture. (3 marks)

End of questions